

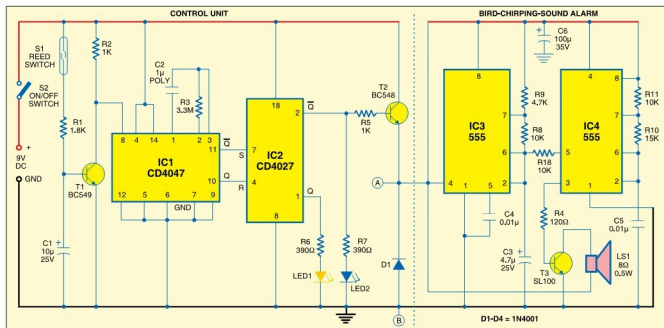


Fig. 1: Alarm circuit that generates bird-chirping sound

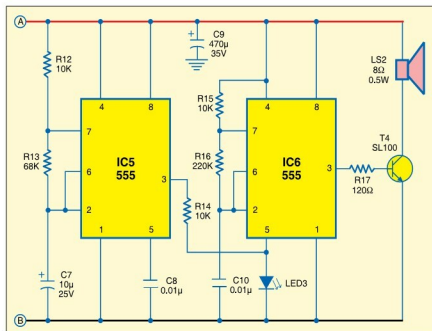


Fig. 2: Alarm circuit that generates police siren tone

monostable multivibrator mode to set and reset IC CD4027 (IC2). The output pulse width of IC1 depends on the values of capacitor C2 and resistor R3 connected to its pins 1, 2 and 3.

Normally, when the door is closed, reed switch S1 is closed, transistor T1 conducts and the monostable multivibrator (IC1) remains in standby mode

with 'low' output at pin 10.

When the door is opened, reed switch S1 gets disconnected, T1 stops conducting and low-to-high pulse at pin 8 of IC1 triggers the monostable and a short-duration positive pulse of about 10 seconds is available as Q output at pin 10. At the same time, complementary output $\bar{Q}$ goes low at

pin 11. The output from IC1 is used to set and reset IC2.

IC2 is a low-power, dual J-K master/slave flip-flop having independent J, K, set, reset and clock inputs. The flip-flops change states on the positive-going transition of the clock pulses. IC2 is wired such that its Q output turns 'high' when reset pin 4 receives a high pulse. When set pin 7 receives a high pulse, Q output goes low and $\bar{Q}$ output goes high. This lights up LED2 and drives transistor T2 (BC548), which enables the alarm circuit.

The output at point A is used to enable the alarm tone generator circuit (on the right side of the dotted line) consisting of two 555 timer ICs marked as IC3 and IC4. The R-C network determines the frequency of the sound produced. The triangular waveform of the astable multivibrator is taken out from the junction of pins 2 and 6 of IC3. This waveform is fed as the control voltage at pin 5 of IC4 through resistor R18. The output received from pin 3 of IC4 is fed to the base of transistor T3 to drive an 8-ohm loudspeaker (LS1), which generates the bird-chirping sound.

For the chirping-sound alarm gen-